

DA-02-T4

Behavioural Data Requirements for Insight Surfacing

DiscountMate | DataBytes | Trimester 2 2026 | Feature 2: Insight Surfacing Power BI Dashboard

1. Purpose

The purpose of this document is to identify the behavioural data required from Feature 1 (Behavioural Analytics) before Feature 2 can develop behavioural insights and persona visualisations in Power BI.

This document is based on the DA-01-T3 — Behavioural Feature Specification, particularly Sections 5 (Behavioural Feature Specification), 6.3 (Data Output), and 8 (Interpretation and Mapping).

The requirements below clarify:

- what behavioural datasets Feature 1 plans to provide;
- which fields are available for dashboard analysis;
- what behavioural and persona insights can be created;
- which requested fields are currently unavailable or deferred;

2. Behavioural Dataset Expected from Feature 1

Feature 1 plans to produce a cleaned, feature-engineered behavioural dataset based on GA4 data exported through BigQuery.

The dataset will contain one row per included user rather than one row per individual event.

Primary source

GA4 raw event export via BigQuery. The specification identifies the following GA4 tables as the primary input:

- Events_YYYYMMDD
- pseudonymous_users_YYYYMMDD

The GA4 data contains multiple events per session and multiple sessions per user.

3. Behavioural Fields Expected

According to Section 6.3 of DA-01-T3, the final dataset supplied to downstream teams will contain:

Field	Expected availability	Description	Potential DA-02 use
user_pseudo_id	Available	Anonymous GA4 client identifier / primary key	User-level behavioural analysis and aggregation
reporting_window_start	Available	Start of aggregation period	Time-based filtering
reporting_window_end	Available	End of aggregation period	Time-based filtering
sessions_count	Available	Distinct sessions per user	Engagement and returning-user analysis
avg_engagement_time_sec	Available	Mean engagement time per session	Engagement analysis
pageviews_per_session	Available	Total pageviews divided by sessions	Browsing intensity
avg_percent_scrolled	Available	Mean scroll depth per pageview	Content engagement
compare_ratio	Available	Proportion of pageviews on /compare	Comparison behaviour
specials_ratio	Available	Proportion of pageviews on /specials	Deal/specials behaviour
mylists_ratio	Available	Proportion of pageviews on /my-lists	Shopping-list behaviour
category_browse_ratio	Available	Proportion of pageviews on /category/{id}	Category browsing
product_detail_ratio	Available	Proportion of pageviews on /product/{id}	Product browsing

Field	Expected availability	Description	Potential DA-02 use
unique_pages_visited	Available	Number of distinct pages visited	Exploration breadth
browsing_entropy	Available	Diversity of navigation paths	Systematic vs exploratory behaviour
days_since_first_visit	Available	User tenure	New vs established users
days_since_last_activity	Available	User recency	Recent vs inactive users
is_included_in_run	Available	Whether the user meets inclusion criteria	Dataset filtering

The 13 behavioural features are grouped into four dimensions:

- Engagement Intensity
- Feature Affinity
- Exploration Style
- Recency and Tenure

4. GA4 Data Used to Produce the Features

Feature 1 is currently relying on GA4 events and parameters rather than custom event names. The specification identifies the following important GA4 information:

GA4 data	Use
session_start	Calculate sessions and user activity
page_view	Calculate pageviews and page-based feature ratios
engagement_time_msec	Calculate engagement time
percent_scrolled	Calculate average scroll depth
page_location	Identify pages such as Compare, Specials, My Lists, Categories and Products
page_referrer	Analyse navigation/referrer chains and browsing behaviour
device.web_info.hostname	Identify production events and exclude non-production data

URL patterns in page_location are particularly important because they are used to identify user interaction with:

- /compare
- /specials
- /my-lists
- /category/{id}
- /product/{id}

5. Data Cleaning Requirements

Before the dataset is supplied to Feature 2, Feature 1's cleaning process is expected to apply the requirements in Section 6.2. The data should:

- include only events from the production hostname discountmate.app;
- remove query parameters from URLs before page-pattern analysis;
- exclude single-event sessions;
- include users with at least one session containing at least 10 seconds of engagement;
- impute zero for applicable count and ratio-based features when the underlying event is missing;
- exclude users with zero total pageviews;
- cap continuous variables at the 99th percentile before scaling.

These cleaning requirements are important because the resulting Power BI insights should be based on the same cleaned population used for clustering.

6. Behavioural Insights That DA-02 Can Create

Based on the currently available features, Feature 2 can develop several behavioural insights in Power BI.

6.1 Engagement insights

Using: *sessions_count*, *avg_engagement_time_sec*, *pageviews_per_session*, *avg_percent_scrolled*

DA-02 could analyse:

- highly engaged vs lightly engaged users;
- average sessions per user;
- average engagement time;
- browsing intensity;
- scroll engagement;
- returning-user behaviour.

6.2 Feature usage

Using: *compare_ratio*, *specials_ratio*, *mylists_ratio*, *category_browse_ratio*, *product_detail_ratio*

DA-02 could identify:

- users who primarily compare products;
- users who focus on specials;
- users who use shopping lists;
- users who browse categories;
- users who spend more time viewing individual products.

This can provide a high-level view of what users actually use on DiscountMate.

6.3 Exploration behaviour

Using: *unique_pages_visited*, *browsing_entropy*

DA-02 could identify:

- highly exploratory users;
- task-directed users;
- users who repeatedly visit a small number of pages;
- users who navigate across many different pages.

6.4 Recency and tenure

Using: *days_since_first_visit*, *days_since_last_activity*

DA-02 could distinguish:

- new users;
- established users;
- recently active users;
- users who have become inactive.

6.5 Persona-specific trends

Once cluster/persona labels are available, DA-02 could compare behavioural characteristics between personas. For example:

Persona hypothesis	Behavioural signals to investigate
Budget-Conscious Shopper	High <i>specials_ratio</i> , high <i>compare_ratio</i>
Family Planner	High <i>mylists_ratio</i> , high <i>sessions_count</i> , longer tenure
Convenience Seeker	Lower engagement time, higher <i>product_detail_ratio</i>
Premium Shopper	Lower <i>specials_ratio</i> , moderate <i>product_detail_ratio</i>
Health Enthusiast	Higher <i>category_browse_ratio</i>

7. Requested Data That Is NOT Currently Available

Several fields that would be useful for Feature 2 were considered in DA-01-T3 but are explicitly deferred. This distinction is important for dashboard planning.

Requested data	Current status	Reason / dependency
Anonymous user ID	Available	user_pseudo_id is included in the output
Event name	Available at source level, but not a main output field	Feature engineering aggregates GA4 events into behavioural features
Page viewed	Available at source level	Used to calculate page-based ratios, but individual pages are not listed as a main final output field
Product ID	Not currently provided as a final behavioural feature	Product-level event data is not specified as a downstream output
Category ID	Partially available	Used in category_browse_ratio; human-readable category mapping depends on Data Engineering
Category name	Not currently available	Requires category ID → category name mapping from Data Engineering
Retailer	Not currently available	Requires outbound-click instrumentation
Search term	Not available	Search functionality / instrumentation does not currently exist
Timestamp	Available at GA4 source level	Aggregated into recency and tenure features rather than supplied as individual behavioural records
Session ID	Available at GA4 source level	Used to calculate session-based features but not listed in the final one-row-per-user output
Persona/cluster label	Planned	Expected after clustering and interpretation
Outbound retailer click	Not available	Requires Outbound_Merchant_Click instrumentation
Retailer preference	Not available	Depends on outbound-click instrumentation
Searches per session	Not available	Search functionality does not currently exist
Filter usage	Not available	Requires platform functionality/instrumentation
Sort usage	Not available	Requires platform functionality/instrumentation
Favourites	Not available	Feature not yet implemented
Price alerts	Not available	Feature not yet implemented
Coupon clicks	Not available	Requires corresponding platform functionality

8. Search Behaviour

Search-related insights are not currently supported by the DA-01-T3 specification. The document explicitly identifies:

- searches_per_session
- filter_usage_rate
- sort_usage_rate

as deferred features. The reason is that the required search/filter/sort functionality and instrumentation do not currently exist.

Therefore, DA-02 cannot currently report:

- most common search terms;
- searches per session;
- searches by persona;
- search-to-product behaviour;
- filter usage;
- sort usage.

9. Shopping List Behaviour

Shopping-list behaviour is partially supported. The current feature specification provides:

- mylists_ratio

This measures the proportion of pageviews occurring on /my-lists. Therefore, Feature 2 can investigate shopping-list page usage.

However, the specification does not currently provide fields such as:

- number of products added to a list;
- products added to lists;
- list creation frequency;
- list completion;
- list-level product categories.

Therefore, DA-02 can currently report My Lists feature engagement, but not detailed shopping-list behaviour.

10. Missing Data / Additional Requirements

The following data would materially improve the behavioural dashboards:

High priority

1. Outbound retailer clicks

This is identified by Feature 1 as the single highest-value future addition. Required event: Outbound_Merchant_Click. Potential fields:

- anonymous user ID;
- timestamp;
- session ID;
- product ID;
- retailer;
- category;
- destination;
- possibly page/source context.

This would allow DA-02 to measure decision behaviour rather than only engagement.

2. Human-readable category mapping

The current category_browse_ratio can identify category browsing using category IDs but Feature 1 requires a Data Engineering mapping to convert category IDs into meaningful category names. This is required for interpretable persona analysis.

3. Product-level interaction data

To create popular-product and frequently-compared-product insights, DA-02 would require product-level identifiers linked to events.

4. Search instrumentation

Search terms and search frequency require search functionality and corresponding GA4 instrumentation.

5. Detailed shopping-list events

For deeper shopping-list analysis, events such as list creation, product addition/removal, and list interaction would be useful.

11. Dependencies

Dependency	Responsible team	Current status	Impact on Feature 2
GA4 extraction (T4)	Feature 1	Planned W5–W6	Dashboard build cannot start until extraction completes
Data cleaning (T5)	Feature 1	Feeds W6 validation run	Determines the cleaned population Feature 2's insights must match
Behavioural feature table	Robin Le / Feature 1	Weekly refresh from W6	Core input for engagement / feature-usage Power BI visuals
Clustering —K-means/DBSCAN (T9)	Shubhpreet Kaur / Feature 1	Target W7–W8	No cluster ID exists until this completes
Persona interpretation (T11)	Feature 1	Following T9	Blocks persona visualisation, needed by W8
Category ID → name mapping	Data Engineering	Unresolved (open decision)	Blocks interpretable category-based insights
Outbound-click instrumentation	Web Dev (escalated by Robin)	Unresolved (open decision)	Blocks retailer and popular-product insights
Persona visualisation in Power BI	Feature 2	Needed by W8	Feature 2's own deliverable

11. Conclusion

The DA-01-T3 specification confirms that Feature 1 will provide Feature 2 with a cleaned, user-level behavioural dataset containing 13 behavioural features across engagement intensity, feature affinity, exploration style, and recency/tenure.

The strongest currently supported Power BI insights are therefore around engagement, feature usage, comparison behaviour, specials browsing, My Lists usage, category browsing, product-detail browsing, exploration style, recency, and tenure.

However, several requested insights are not currently supported by the T2 data contract. In particular, search terms, detailed product-level popularity, retailer preference, outbound retailer clicks, and detailed shopping-list activity require additional instrumentation, platform functionality, or catalogue data.

Source: DA-01-T3 — Behavioural Feature Specification v1.0 (Robin Le)